



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2009

Yearly Examination

YEAR 11

Mathematics (2 unit) & Extension Continuers

Instructions:

- Each Question is to be returned in a separate booklet.
- **Question 1 & 2 are to be collected after 60 minutes** at which time the 2 unit Mathematics students will be dismissed.
- **Question 3 & 4 are to be collected after 105 minutes.**
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise stated.
- Full marks may not be awarded for careless or badly arranged work.

Examiner: *A Fuller*

(Use a SEPARATE writing booklet)

Question 1 (28 marks)

(a) Find the value of $\log_3 9$.

1

(b) Solve the following for x :

6

(i) $\log_6 x = 3$

(ii) $\log_x 3 = -1$

(iii) $2^{2x+1} = \frac{1}{16}$

(iv) $x^2 + 3x - 18 = 0$

(c) Differentiate the following with respect to x :

7

(i) $2x + 5$

(ii) $\frac{1}{2x + 5}$

(iii) $\frac{2x + 5}{x}$

(iv) $\frac{x}{2x + 5}$

- (d) Sketch the following on separate axes showing any intercepts with the co-ordinate axes and any asymptotes:

4

(i) $y = \frac{1}{x} + 1$

(ii) $y = \log_2(x + 1)$

- (e) Write $2x^2 - 7x - 4$ in the form $a(x + 2)^2 + b(x + 2) + c$.

3

- (f) Consider the arithmetic series: $-1 + 3 + 7 + 11 + 15 + \dots$

4

- (i) Which term of the series is 391?

- (ii) Hence, find the sum up to the term which is 391.

- (g) If $f(x) = x^3 - 3x^2 - 6x$. Evaluate:

3

(i) $f(-2)$

(ii) $f'(2)$

(Use a SEPARATE writing booklet)

Question 2 (28 marks)

(a) Consider the geometric series: $27 + 18 + 12 + 8 + \dots$

2

(i) Explain why the series has a limiting sum.

(ii) Find the limiting sum of the series.

(b) Find the co-ordinates of the focus and the equation of the directrix of the parabola $y = \frac{x^2}{4} - 1$.

3

(c) State whether the following functions are odd, even, or neither:

3

(i) $f(x) = x^6 + 10$

(ii) $f(x) = \frac{x^2}{2 - x}$

(iii) $f(x) = \log_{10} 2^x$

(d) Find using first principles the derivative $f'(x)$ given that $f(x) = x^3$.

2

(e) Let $\log_5 3 = a$ and $\log_5 2 = b$.

4

(i) Find the following in terms of a and b :

(α) $\log_5 6$

(β) $\log_5 \left(\frac{1}{4}\right)$

(ii) Evaluate 5^{2a} .

(f) Find the value(s) of k for which $x^2 - kx + 4 = 0$ has:

5

(i) one root equal to -1

(ii) real roots

(iii) one root double the other.

(g) Find the domain and the range of the following:

4

(i) $y = \sqrt{1 - x}$

(ii) $y = \sqrt{1 - x^2}$

- (h) Caleb plans to deposit an amount of money into an account which will pay him 1% interest each month on the balance of his account at the time.

Immediately after each interest payment is made, Caleb plans to withdraw \$1000.

Let his deposit be $\$D$.

- (i) Show that when he has made his second withdrawal, the balance of his account will be $\$[D(1.01)^2 - 1000(1 + 1.01)]$.

- (ii) Caleb wants his deposit to be sufficient to be able to make withdrawals for 10 years. Find, to the nearest \$100, what his deposit must be.

$$(a) \log_3 9 = 2 \quad [1]$$

$$(b) (i) \log_6 x = 3$$

$$x = 6^3$$

$$x = 216 \quad [1]$$

$$(ii) \log_x 3 = -1$$

$$3 = x^{-1}$$

$$x = \frac{1}{3} \quad [1]$$

$$(iii) 2^{2x+1} = \frac{1}{16}$$

$$2^{2x+1} = 2^{-4}$$

$$2x+1 = -4$$

$$x = -2\frac{1}{2} \quad [2]$$

$$(iv) x^2 + 3x - 18 = 0$$

$$(x+6)(x-3) = 0$$

$$x = -6 \text{ or } 3 \quad [2]$$

$$(c) (i) \frac{d}{dx}(2x+5) = 2 \quad [1]$$

$$(ii) \frac{d}{dx}\left(\frac{1}{2x+5}\right) = \frac{d}{dx}((2x+5)^{-1})$$

$$= -1 \times (2x+5)^{-2} \times 2$$

$$= \frac{-2}{(2x+5)^2} \quad [2]$$

$$(iii) \frac{d}{dx}\left(\frac{2x+5}{x}\right) = \frac{d}{dx}\left(2 + \frac{5}{x}\right)$$

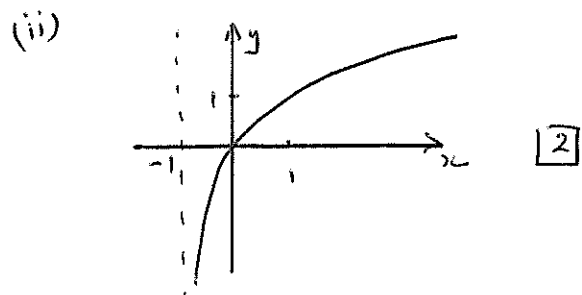
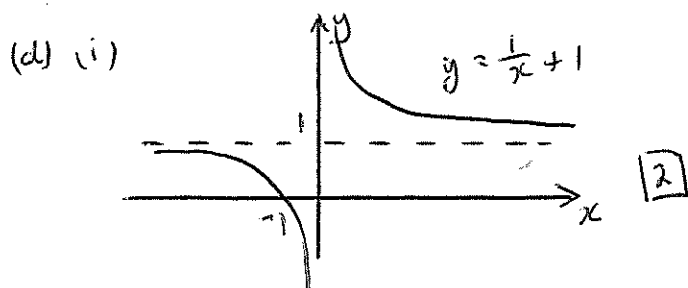
$$= 5x^{-1} - 1x^{-2}$$

$$= \frac{-5}{x^2} \quad [2]$$

$$(iv) \frac{d}{dx}\left(\frac{x}{2x+5}\right) = \frac{(2x+5) \cdot 1 - x \cdot 2}{(2x+5)^2}$$

$$= \frac{2x+5-2x}{(2x+5)^2}$$

$$= \frac{5}{(2x+5)^2} \quad [2]$$



(e) $2x^2 - 7x - 4 = a(x+2)^2 + b(x+2) + c$

Equating leading coefficients $a = 2$

Let $x = -2$: $8 + 14 - 4 = c$

$c = 18$

Let $x = 0$: $-4 = 2 \times 4 + b \times 2 + 18$

$$-4 = 2b + 26$$

$$2b = -30$$

$$b = -15 \quad [3]$$

$$2x^2 - 7x - 4 = 2(x+2)^2 - 15(x+2) + 18$$

(f) $a = -1$

$d = 4$

(i) $T_n = -1 + (n-1) \cdot 4 = 391$

$$4(n-1) = 392$$

$$n-1 = 98$$

$$n = 99 \quad [2]$$

(ii) $S_{99} = \frac{99}{2}(-1 + 391)$

$$= \frac{99}{2} \times 390$$

$$= 19305 \quad [2]$$

(g) (i) $f(-2) = -8 - 12 + 12$

$$= -8 \quad [1]$$

(ii) $f'(x) = 3x^2 - 6x - 6$

$$f'(2) = 12 - 12 - 6$$

$$= -6 \quad [2]$$

28

Question 2.

$$f(-x) = \frac{(-x)^2}{2 - (-x)}$$
$$= \frac{x^2}{2 + x}$$

a) If $|r| < 1$, the sum of
① the series is limited to
some finite number ①

$\therefore f(x)$ is neither ①

ii) $a = 27$ $r = \frac{2}{3}$

iii) $f(x) = \log_{10} 2^x = x \log_{10} 2$

$$S_{\infty} = \frac{27}{1 - \frac{2}{3}}$$

$$= 81$$

①

$$f(-x) = \log_{10} 2^{-x}$$

$$= -x \log_{10} 2$$

$$= -f(x)$$

b) $y = \frac{x^2}{4} - 1$

$\therefore f(x)$ is odd. ①

$$4y = x^2 - 4$$

$$x^2 = 4y + 4$$

$$x^2 = 4(y + 1)$$

d) $f'(x) = \frac{f(x+h) - f(x)}{h}$

$$= \frac{(x+h)^3 - x^3}{h}$$

$\therefore h = 0, k = -1, a = 1$ ①

focus: $(0, 0)$

①

$$(x+h)^3 = (x^2 + 2xh + h^2)(x+h)$$
$$= x^3 + x^2h + 2x^2h + 2xh^2$$
$$+ h^2x + h^3$$

$$= x^3 + 3x^2h + 3xh^2 + h^3$$

directrix: $y = -1 - 1$

$$y = -2$$

①

$$= \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$= \frac{3x^2h + 3xh^2 + h^3}{h}$$

$$= \cancel{h} (3x^2 + 3xh + h^2)$$

c) i) $f(x) = x^6 + 10$

$$f(-x) = (-x)^6 + 10$$

$$= x^6 + 10$$

$\therefore f(x)$ is even ①

$$\lim_{h \rightarrow 0} = 3x^2$$

②

ii) $f(x) = \frac{x^2}{2-x}$

$$h \rightarrow 0$$

$$\therefore f'(x) = 3x^2$$

$$e) \log_5 3 = a$$

$$\log_5 2 = b$$

$$i) \alpha) \log_5 6 = \log_5 3 + \log_5 2$$

$$= a + b \quad (1)$$

$$\beta) \log_5 (1/4) = \log_5 1 - \log_5 4$$

$$= 0 - \log_5 2^2$$

$$= 0 - 2\log_5 2$$

$$= -2b \quad (1)$$

$$ii) 5^{2a} = (5^a)^2$$

$$= 3^2$$

$$= 9 \quad (2)$$

$$f) x^2 - kx + 4 = 0$$

$$i) (-1)^2 + k + 4 = 0$$

$$k = -5 \quad (1)$$

$$ii) \Delta > 0$$

$$\therefore b^2 - 4ac > 0$$

$$b^2 - 4ac > 0$$

$$(-k)^2 - 4 \times 1 \times 4 > 0$$

$$k^2 - 16 > 0$$

$$k^2 > 16$$

$$k > 4, k < -4 \quad (2)$$

$$iii) \alpha + \beta = -b/a$$

$$\alpha + 2\alpha = k$$

$$3\alpha = k$$

$$\alpha \times 2\alpha = 4$$

$$2\alpha^2 = 4$$

$$\alpha^2 = 2$$

$$\alpha = \pm \sqrt{2}$$

$$\therefore k = 3\sqrt{2}, -3\sqrt{2} \quad (2)$$

$$g) i) D: x \leq 1 \quad (1)$$

$$R: y \geq 0 \quad (1)$$

$$ii) D: -1 \leq x \leq 1 \quad (1)$$

$$R: 0 \leq y \leq 1 \quad (1)$$

h) i) After 1 month

$$A_1 = (\$D \times 1.01) - 1000$$

After 2nd month

$$A_2 = (A_1 \times 1.01) - 1000$$

$$= [(\$D \times 1.01) - 1000] \times 1.01 - 1000$$

$$= \$D \times (1.01)^2 - 1000 \times 1.01 - 1000$$

$$= \$[D(1.01)^2 - 1000(1.01 + 1)]$$

$$= \$[D(1.01)^2 - 1000(1 + 1.01)] \quad (2)$$

$$D(1.01)^{120} - 1000[1.01^{120} + 1.01^{119} + 1.01^{118} + \dots + 1.01 + 1] \geq 0$$

this part is a series

$$a = 1, r = 1.01 \quad n = 120$$

$$S_n = \frac{1(1.01^{120} - 1)}{1.01 - 1}$$

$$D(1.01)^{120} - 1000 \times \frac{1(1.01^{120} - 1)}{1.01 - 1} > 0$$

$$D(1.01)^{120} > 1000 \times \frac{1(1.01^{120} - 1)}{1.01 - 1}$$

$$D > \frac{1000 \times (1.01^{120} - 1)}{1.01 - 1}$$

$$(1.01)^{120}$$

$$D > \$69,700 \text{ (nearest \$100)}$$